

Symmetric functions and their relation to free field vertex algebras

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Joint work with

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Outline

- 1 Universal vs minimal vertex operator algebras
- 2 Free field realisations
- 3 Symmetric polynomials
- 4 Classifying simple modules

Universal vs minimal vertex operator algebras

Def: Universal vertex operator algebra

Freely generated vertex operator algebras. Come in families parametrised by central elements.

In the examples to be considered here:

- simple for generic values of central elements – proper ideals trivial
- non-trivial proper ideals exist for special values of central elements
- **minimal models**: simple quotients at special values of central elements

Examples

Universal Virasoro: generator: T , central element c ,

$$\text{relation: } T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w},$$

$$\text{non-simple} \Leftrightarrow c = 1 - 6\frac{(p-q)^2}{pq}, \quad p, q \in \mathbb{Z}_{\geq 2}, \quad \gcd\{p, q\} = 1.$$

Universal $\widehat{\mathfrak{sl}}(2)$: generators: basis of $\mathfrak{sl}(2)$, central element k ,

$$\text{relations: } X(z)Y(w) \sim \frac{k(X|Y)}{(z-w)^2} + \frac{[X, Y](w)}{z-w}, \quad X, Y \in \mathfrak{sl}(2),$$

$$\text{non-simple} \Leftrightarrow k = -2 + \frac{p}{q}, \quad p \in \mathbb{Z}_{\geq 2}, \quad q \in \mathbb{Z}_{\geq 1}, \quad \gcd\{p, q\} = 1.$$

Universal $N = 1$ Virasoro: generators: T, G , central element c ,

$$\text{relations: } T(z)G(w) \sim \frac{\frac{3}{2}G(w)}{(z-w)^2} + \frac{\partial G(w)}{z-w},$$

$$G(z)G(w) \sim \frac{c/3}{(z-w)^3} + \frac{2T(w)}{z-w},$$

$$\text{non-simple} \Leftrightarrow c = \frac{3}{2} - 3\frac{(p-q)^2}{pq}, \quad p, q \in \mathbb{Z}_{\geq 2}, \quad p+q \in 2\mathbb{Z}, \\ \gcd\{p, \frac{p+q}{2}\} = 1.$$

Universal vs minimal modules

- Module theory of universal vertex operator algebras very hard.
But, contains easy families: Verma modules
- Universal VOA module M :
 M is a minimal model module $\Leftrightarrow$ Ideal in universal VOA acts trivially.
- Minimal model module theory strongly constrained

Example: Virasoro universal vs minimal modules

Universal: Any module with conformal weights bounded below, such as Verma modules, staggered modules and *many* more.

Minimal: $\frac{1}{2}(p-1)(q-1)$ simple modules, all modules semisimple.

Free field vertex operator algebras

Def: Free field algebra

Finitely generated vertex algebra. Operator product expansions of generators are proportional to identity.

Standard examples:

Neutral boson NB: $a(z)a(w) \sim \frac{1}{(z-w)^2}$

Neutral fermion NF: $\psi(z)\psi(w) \sim \frac{1}{z-w}$

Charged bosons CB: $\beta(z)\gamma(w) \sim \frac{-1}{z-w}, \quad \beta(z)\beta(w) \sim 0 \sim \gamma(z)\gamma(w)$

Charged fermions CF: $b(z)c(w) \sim \frac{1}{z-w}, \quad b(z)b(w) \sim 0 \sim c(z)c(w)$

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Free field realisations

- Universal vertex operator algebras $\xrightarrow{\text{embedding}}$ free field algebras.
 $\Rightarrow$ Free field modules also universal modules. [Feigin-Fuchs; Wakimoto; Kac-Frenkel; Kac-Wakimoto]
- Free field module theory is easy.
- Ideals easy to construct in free field algebra.
 $\Rightarrow$ Can evaluate ideals on free field modules.

Examples:

- Virasoro $\hookrightarrow \text{NB}$
- $\widehat{\mathfrak{g}} \hookrightarrow \text{NB}^{\otimes \text{rk } \mathfrak{g}} \otimes \text{CB}^{\otimes \# \text{ pos. roots}}$
 $\Rightarrow \widehat{\mathfrak{sl}}(2) \hookrightarrow \text{NB} \otimes \text{CB}$
- $N = 1$ Virasoro $\hookrightarrow \text{NB} \otimes \text{NF}$
- $\widehat{\mathfrak{osp}}(1|2) \hookrightarrow \text{NB} \otimes \text{CB} \otimes \text{CF}$

Free field realisations

Universal vertex operator algebra V , free field algebra F :

$$V \hookrightarrow F = \mathbf{NB}^{\otimes r} \otimes \mathbf{CB}^{\otimes s} \otimes \mathbf{NF}^{\otimes t} \otimes \mathbf{CF}^{\otimes u}$$

- For every factor $\mathbf{NB}$ in $F \Rightarrow$ exists family of vertex operators $V_\lambda(z), \lambda \in \mathbb{C}$.
- Vertex operators define maps between free field modules
- Screening field: $S(z) = (\text{bosons and fermions}) V_\lambda(z)$, such that operator product expansion of $S(z)$ with V is total derivative
- Screening operators: Residue of $S(z)$. Commutes with V .
 $\Rightarrow$ Residue of $S(z)$ defines V -module homomorphism.
[Dotsenko-Fateev; Tsuchiya-Kanie]
- Ideal of $V \subset F$ can be constructed as image of screening operators [Felder; Bernard-Felder; Bouwknegt-McCarthy-Pilch]

Examples of screening operators

- Virasoro $\hookrightarrow$ NB: $c = 1 - 6 \frac{(p-q)^2}{pq}$

$$S_+(z) = V_{\sqrt{\frac{2q}{p}}}(z), \quad S_-(z) = V_{-\sqrt{\frac{2p}{q}}}(z)$$

- $\widehat{\mathfrak{sl}}(2) \hookrightarrow \text{NB} \otimes \text{CB}$: $k = -2 + \frac{p}{q}$

$$S(z) = \beta(z) V_{-\sqrt{\frac{2q}{p}}}(z)$$

- $N=1$ Virasoro $\hookrightarrow \text{NB} \otimes \text{NF}$: $c = \frac{3}{2} - 3 \frac{(p-q)^2}{pq}$

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- $\widehat{\mathfrak{osp}}(1|2) \hookrightarrow \text{NB} \otimes \text{CB} \otimes \text{CF}$: $k = -\frac{3}{2} + \frac{p}{2q}$

$$S(z) = (\beta(z)c(z) - b(z)) V_{-\sqrt{\frac{q}{p}}}(z)$$

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The appearance of symmetric polynomials

Ideal of V in F can be constructed as image of screening operators

$$S^{[r]} = \int S(z_1) \cdots S(z_r) dz_1 \cdots dz_r.$$

Product of screening fields expands as

$$\begin{aligned} S(z_1) \cdots S(z_r) &= (\text{bosons and fermions}) V_\lambda(z_1) \cdots V_\lambda(z_r) \\ &= \underbrace{(\text{bosons and fermions})}_{(\text{skew}) \text{ symmetric}} \underbrace{\prod_{1 \leq i < j \leq r} (z_i - z_j)^{\lambda^2}}_{(\text{Vandermonde det})^{\lambda^2}} \underbrace{: V_\lambda(z_1) \cdots V_\lambda(z_r) :}_{\text{symmetric}}. \end{aligned}$$

Vandermonde determinant defines inner product for symmetric polynomials

$$\langle f, g \rangle = \int \prod_{1 \leq i \neq j \leq r} (z_i - z_j)^{\lambda^2/2} z_i^{(r-2)\lambda^2/2} f(z^{-1}) g(z) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}.$$

Exists orthogonal basis called **Jack polynomials**.

Symmetric polynomials

Ring of symmetric polynomials $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathcal{S}_r}$.

Admits many bases such as *monomial symmetric polynomials*

$$m_{\lambda}(z) = \sum_{\sigma} z_1^{\sigma_1} \cdots z_r^{\sigma_r},$$

$\lambda = [\lambda_1, \dots, \lambda_r]$ partition, σ distinct permutations of λ .

Examples:

- $m_{[1]}(z_1, z_2) = m_{[1,0]}(z_1, z_2) = z_1 + z_2$
- $m_{[2,2]}(z_1, z_2) = z_1^2 z_2^2$
- $m_{[3,2,1]}(z_1, z_2, z_3) = z_1^3 z_2^2 z_3 + z_1^3 z_2 z_3^2 + z_1^2 z_2^3 z_3 + z_1^2 z_2 z_3^3 + z_1 z_2^3 z_3^2 + z_1 z_2^2 z_3^3$
- $m_{[3,2,1]}(z_1, z_2) = 0$

Jack symmetric polynomials

Jack symmetric polynomials $P_\lambda^t(z)$ are basis of Λ_r characterised by

① Upper triangular decomposition:

$$P_\lambda^t(z) = m_\lambda(z) + \sum_{\mu < \lambda} c(t; \lambda, \mu) m_\mu(z),$$

where λ, μ are partitions and $<$ is dominance partial ordering.

② Orthogonal:

$$\langle f, g \rangle^t = \int \prod_{1 \leq i \neq j \leq r} (z_i - z_j)^{1/t} \prod_{i=1}^r z_i^{(r-1)/t} f(z^{-1}) g(z) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

$\Rightarrow$ Screening operators distinguish Jack polynomials as **natural basis**.

Classifying simple minimal model modules

Evaluating ideals on free field modules $\Rightarrow$ Polynomial constraints on weights of (relaxed) highest weight vectors

$\Rightarrow$ Simple module classification through classification of weights

[Feigin-Ooguri-Nakanishi, Wang; Ridout-SW]

Virasoro: $c = 1 - 6 \frac{(p-q)^2}{pq}$

$$\left\langle \mathbf{P}_{[(q-1)^{p-1}]}^{p/q}, \prod_{i=1}^{p-1} (1 + z_i)^{\sqrt{\frac{2q}{p}} \tau} \right\rangle^{p/q} \propto \prod_{r,s} (h_\tau - h_{r,s}),$$

τ neutral boson weight; h_τ conformal weight; $r, s \in$ Kac table

Classifying simple minimal model modules

Evaluating ideals on free field modules $\Rightarrow$ Polynomial constraints on weights of (relaxed) highest weight vectors

$\Rightarrow$ Simple module classification through classification of weights

[Adamović-Milas; Ridout-SW]

$$\widehat{\mathfrak{sl}}(2): k = -2 + \frac{p}{q}$$

$$\sum_{n=0}^{p-1} (-1)^n \binom{p-2-n+\sigma}{p-1-n} \sum_{m=0}^n \binom{p-1-m}{n-m} \\ \times \left\langle \mathbf{P}_{[q^m, (q-1)^{p-1-m}]}^{p/q}, \prod_{i=1}^{p-1} (1+z_i)^{-\sqrt{\frac{2q}{p}}\tau-1} \right\rangle^{p/q} \\ \propto g_p(\lambda_{\tau, \sigma}, h_{\tau}) \prod_{r,s} (h_{\tau} - h_{r,s}),$$

τ neutral boson weight, σ charged boson weight; $\lambda_{\tau, \sigma}$ $\widehat{\mathfrak{sl}}(2)$ weight, h_{τ} conformal weight; $r, s \in$ Kac table

Classifying simple minimal model modules

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[Adamović, Milas; Ridout-SW]

$N = 1$ Virasoro: $c = \frac{3}{2} - 3 \frac{(p-q)^2}{pq}$

$$\begin{aligned} \text{NS: } & \left\langle P_{\delta}^{-3}, \prod_{i=1}^{p-1} (1+z_i)^{\sqrt{\frac{q}{p}}\tau} \right\rangle^{2p/(q-p)} \propto \prod_{r,s} (h_{\tau} - h_{r,s}^{(\text{NS})}) \\ \text{R: } & \left\langle \sum_{\lambda} c_{\lambda} P_{\lambda}^{-3}, \prod_{i=1}^{p-1} (1+z_i)^{\sqrt{\frac{q}{p}}\tau} \right\rangle^{2p/(q-p)} \propto \prod_{r,s} (h_{\tau} - h_{r,s}^{(\text{R})}) \end{aligned}$$

τ neutral boson weight; h_{τ} conformal weight; $r, s \in \text{Kac table}$

Why P_δ^{-3} ?

P_δ^{-3} contribution due to fermions in screening operator $\psi(z)V_{\sqrt{\frac{q}{p}}}(z)$

$$\langle \psi(z_1) \cdots \psi(z_{2n}) \rangle = \frac{1}{z_1 - z_2} \frac{1}{z_3 - z_4} \cdots \frac{1}{z_{2n-1} - z_{2n}} + \text{skew symm}$$

$\Rightarrow \prod_{1 \leq i < j \leq 2n} (z_i - z_j) \langle \psi(z_1) \cdots \psi(z_{2n}) \rangle$ is

- 1) symmetric
- 2) vanishes if $z_1 = z_2 = z_3$

Theorem: [Feigin, Jimbo, Miwa, Mukhin]

Ideal of symmetric polynomials f , where $f = 0$ if $z_1 = z_2 = z_3$, is spanned by Jack polynomials P_λ^{-3} with $\lambda_i - \lambda_{i+2} \geq 2$.

Fermions distinguish **wrong** family of Jack polynomials.

Big picture

$$\widehat{\mathfrak{sl}}(2)$$

Virasoro

$$\widehat{\mathfrak{sl}}(n)$$

W_n algebras

$$\widehat{\mathfrak{osp}}(1|2)$$

$N = 1$ Virasoro

$$\widehat{\mathfrak{sl}}(2|1)$$

Hamiltonian reduction $\rightarrow$

$N = 2$ Virasoro

$$\widehat{\mathfrak{osp}}(3|2)$$

$N = 3$ Virasoro

$$\widehat{\mathfrak{psl}}(2|2)$$

$N = 4$ Virasoro

$$\widehat{\mathfrak{d}}(2|1; \alpha)$$

Goal: Study all minimal modules uniformly.

Big picture

$$\widehat{\mathfrak{sl}}(2) \quad \checkmark$$

$$\widehat{\mathfrak{sl}}(n)$$

$$\widehat{\mathfrak{osp}}(1|2) \quad (\checkmark)$$

$$\widehat{\mathfrak{sl}}(2|1)$$

$$\widehat{\mathfrak{osp}}(3|2)$$

$$\widehat{\mathfrak{psl}}(2|2)$$

$$\widehat{\mathfrak{d}}(2|1; \alpha)$$

Hamiltonian reduction $\rightarrow$

Virasoro $\checkmark$

W_n algebras

$N = 1$ Virasoro $\checkmark$

$N = 2$ Virasoro

$N = 3$ Virasoro

$N = 4$ Virasoro

Goal: Study all minimal modules uniformly.

Thank you for coming!